WEEK-2 ASSIGNMENT:

In this assignment, we require:

- 4 LEDs
- 4 Pushbuttons
- 1 LCD
- 1 Buzzer

As soon as, the simulation starts, the user is displayed “Game Starts” on LCD.

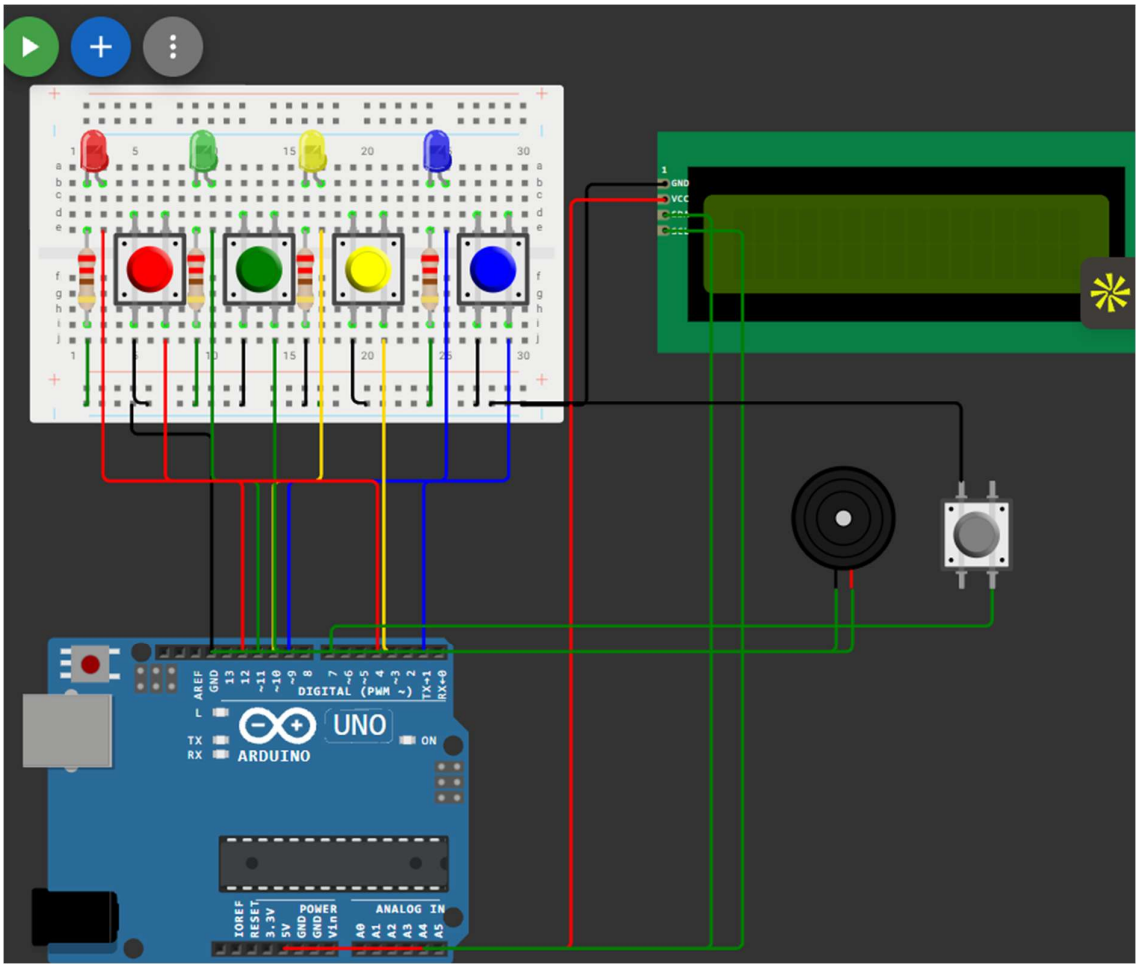
Then we have states 0,1,2 mapped to IDLE,GAME RUNNING,GAME OVER respectively.

0,1,2,3 mapped to red, green, yellow and blue LEDs.

Game starts with **IDLE** state: the system waits for the user to press the start button. As soon as button is pressed, StartTime is recorded and:

State switches to **GAME RUNNING** state: where current glowing LED is turned off and a random LED is lit using the random() function. it detects if the button pressed matches the LED, score is incremented by 1 and a success tone is played by the buzzer while an error tone if button does not match the LED. When time reaches the game duration:

State switches to **GAME OVER** state: where final score is displayed by LCD and state switches back to IDLE resetting all the conditions.



Link to Project: https://wokwi.com/projects/452409517759309825